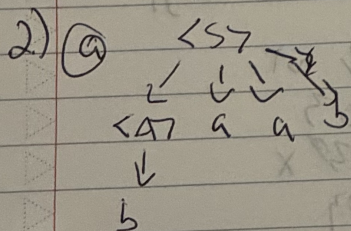


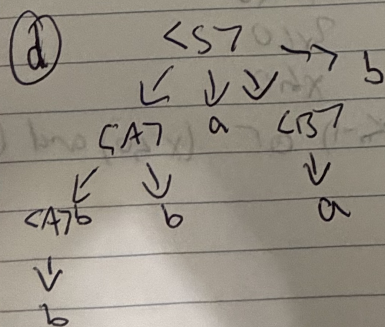
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CSIT 313 Assignment 2

- 1) $S \rightarrow \langle A \rangle \langle B \rangle \langle C \rangle$
 $\langle A \rangle \rightarrow a \langle A \rangle | a$
 $\langle B \rangle \rightarrow b \langle B \rangle | b$
 $\langle C \rangle \rightarrow c \langle C \rangle | c$

This means that S is made from $\langle A \rangle \langle B \rangle \langle C \rangle$.
 $\langle A \rangle$ can be a or $a \langle A \rangle$ so it is a^n . Same goes for $\langle B \rangle$ and $\langle C \rangle$, $\langle A \rangle \rightarrow a \langle A \rangle | a$ is an infinite loop as it can multiply by itself. Or it can stop with the other outcome a . $\langle A \rangle$ is a parent and a is a child.



(b) $\langle S \rangle$
 $\langle S \rangle$ cannot exist because it has to end with minimum aab .



(c) $\langle C \rangle$ cannot be created because there is not another outcome that creates an $\langle S \rangle$.

3) (a) $a = 2^b (b-1) - 1 \{a \geq 3\}$ weakest precondition = $(b \geq 3)$

$$2b - 2 - 1 \geq 3$$

$$2b - 3 \geq 3$$

$$2b \geq 6$$

$$b \geq 3$$

⑤ $b = \lfloor (c+10)/3 \rfloor \wedge b < 8$ weakest precondition: $(c \leq 8)$
 $\frac{(c+10)}{3} < 8$

$(c+10) < 24$
 $c < 14$

⑥ $a = a + 2(b-1) \{a \geq 5\}$ weakest precondition: $(a \geq 2b - 6)$
 $a + 2b - 2 \geq 5$
 $a + 2b \geq 7$

⑦ if $(x \leq y)$	$(x \leq y) \text{ and } (x \leq -1)$	$(x \leq 0) \text{ and } (x \geq y)$
$x = x + 1$		
else	$\downarrow$	$\downarrow$ not
$x = 3 * x$	$\{x \leq -1\}$	$\{x \leq 0\}$
$\{x \leq 0\}$	$x = x + 1$	$x = 2 * x$
	$\{x \leq 0\}$	$\{x \leq 0\}$

$x + 1 \leq 0$
 $x \leq -1$

$3 * x \leq 0$
 $x \leq 0$
 $x \leq 0$

weakest precondition $\equiv (x \geq y) \text{ and } (x \leq -1) \text{ or } (x \leq 0) \text{ and } (x \geq y)$